Diego Pilares Gallego

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Education

Oxford Brookes University — Oxford, UK
MSc in Motorsport Engineering. Merit — September 2024 – November 2025

Thesis: Hydrogen-Fuelled LTC/CAI-IC Engine Control Strategy Optimisation
Relevant Modules: • Advanced Vehicle Dynamics • Composite Design and Impact Modelling

Universidad Francisco de Vitoria – Motor & Sport Institute — Alcorcon, Spain
Beng in Automotive Engineering — September 2019 – January 2025

Thesis: Mono-Nut Carbon Fibre Wheel CAD and FEA Validation for Formula Student
Relevant Modules: • Suspension • Transmission and Gearbox • Powertrain • Machines and Mechanisms Theory • Design, Prototyping and Testing

Experience

CT Ingenieros – HORSE Powertrain — Valladolid, Spain
Test Methodology Engineer — May 2026 – Present

- Involved in the development of testing methodology for chassis dyno, robotized chassis dyno and PEMS.
- Development of AVL Concerto tools.
- Functions developed mainly in AVL and Horiba environments.

Oxford Brookes Racing — Oxford, UK
Formula Student Team Member – Suspension & Structures — September 2024 – May 2025

- Designed brake rotors and hub components using SolidWorks and HyperMesh (OptiStruct).
- Applied topology optimisation to reduce mass while maintaining performance and reliability.

NTT DATA — Madrid, Spain
Industrial Engineer Placement – Automotive Department — February 2024 – September 2024

- Developed corrective measures and new functionalities for BMW online platforms across multiple international markets.
- Produced technical documentation and training materials.
- Analysed and resolved functional and data-related issues.
- Designed layouts for Hyundai Spain internal applications.

UFV Racing — Alcorcon, Spain
Formula Student Team Member – Electronics, Systems & Testing — September 2019 – July 2021

- Designed CAD, wiring diagrams, PCB layouts, supported AIM PDM integration and engine mapping.
- Installed powertrain, suspension, steering, and brake systems.

Skills & Interests

Simulation: AVL Concerto, AVL VSM, AVL Drive Race, MoTeC i2, GT-Suite, MSC Adams, MATLAB/Simulink, Multisim
CAD, FEA & CFD: CATIA (V5 & 3DX), SolidWorks, Star CCM+, XFLR5, Femap, HyperMesh
Productivity: Microsoft Office, Google Workspace, Jira
Language: Spanish (Native), English (C1, Professional working proficiency), German (A1, Basic)
Interests: Motorsports, Go-karting, Retro electronics restoration, Hands-on maintenance

Projects

MSc Dissertation - Hydrogen-Fuelled LTC/CAI-IC Engine Control Strategy Optimisation Oxford Brookes University
January 2025 – September 2025

- Developed engine control strategy for hydrogen LTC/CAI engine, combining 1D engine simulation with Machine-Learning optimization.
- Built and calibrated a full 1D engine model in GT-Suite, then applied DOE and ML algorithms to optimise control parameters, maximized thermal efficiency while minimising NOx emissions.
- The work sits at the intersection of alternative fuels, engine simulation and data-driven control, directly relevant to the industry's push towards zero-emission powertrains.

LMGT3 DrivAer Front Aero Device CAD and CFD Validation Oxford Brookes University
October 2024 – December 2024

- End-to-end design and CFD validation of front aerodynamic devices for a DrivAer model, following LMGT3 competition regulations.
- Geometry's CAD developed in SolidWorks and validated through full CFD simulation in Star CCM+, assessing downforce, drag and flow interaction with the front axle and full aerodynamic package developed by other team members.
- Included development of a wind tunnel validation methodology to support physical testing correlation. Demonstrates a complete aerodynamic development workflow, from concept to simulation-backed performance assessment.

Suspension Dynamics and Ride Optimisation Oxford Brookes University
October 2024 – December 2024

- Built bicycle, twin-track and 1 to 4 DoF models in MSC Adams, progressing from simplified representations to full vehicle dynamics simulations. Analysed cornering response, vertical load transfer and ride comfort across a range of operating conditions.
- Results informed suspension geometry and setup recommendations aimed at improving both driver feedback and mechanical grip.

Additional Information

Portfolio: Complete project documentation and technical work available at dpilares01.github.io
Currently taking certified CATIA 3DX courses.
Currently developing a python telemetry analysis tool as a personal learning project.

Work Authorisation: EU citizen with unrestricted right to work.